MILITARY SPECIFICATION

REPEATER SET, RADIO, POD
AN/ARQ-49

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for Repeater Set, Radio, Pod AN/ARQ-49 herein referred to as the pod. The pod will be used to relay UHF communication signals and Navy Tactical Data System (NTDS) Link 11 signals.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-P-35	Passivation Treatments for Corrosion - Resisting Steel
QQ-P-416	Plating, Cadmium (Electrodeposited)

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MIL-P-116	Preservation-Packaging, Methods of
MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems
MIL-W-5088	Wiring; Aircraft, Selection and Installation of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associate Equipment
MIL-C-5541	Chemical Conversion Coatings Aluminum and Aluminum Alloys
MIL-C-6021	Casting, Classification and Inspection of
MIL-M-7793	Meter, Time Totalizing
MIL-A-8591	Airborne Stores, Associated Suspension Lugs, and Aircraft-Store Interface (Carriage Phase), General Design Criteria for
MIL-P-15024/2	Plate, Identification Unit or Plug-in Assembly
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-F-18264	Finishes, Organic, Weapons System, Application and Control of
MIL-T-23103	Thermal Performance Evaluation Airborne Electronic Equipment, General Requirement for
MIL-P-23377	Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, for Aerospace Applications

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STANDARDS

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MIL-STD-130	Identification Markings of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-756	Reliability Prediction
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-889	Dissimilar Metals
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2084	Maintainability of Avionics Systems and Equipment, General Requirements

HANDBOOKS

MILITARY

MIL-HDBK-217	Reliability Prediction of Electronic Equipment

2.1.2 Documents and drawings. The following documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DOCUMENTS

NAVAL AIR SYSTEMS COMMAND

AS-4613	Application and Derating Requirements for Electronic Components, General Specification for
WS-6536	Process Specification Procedures and Requirements for Preparation and Soldering of Electrical Connections
NAC R&M-STD-R00301.1	Manufacturing Burn-In/Screening; Unit/System Level

DRAWING

NAVAL AIR SYSTEMS COMMAND

1536AS100	Repeater Set, Radio, Pod AN/ARQ-49

(Copies of specifications, standards, handbooks, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The AN/ARQ-49 shall be in accordance with Drawing 1536AS100. The baseline configuration shall consist of four Radio RT-1250A/ARC Receiver-Transmitters to provide two voice channels and two radio RT-1486/A Receiver-Transmitters to provide a Link 11 channel. These units shall be contained in a 15-foot pod which shall have incorporated a ram air turbine (RAT) for power, and fans for cooling. RAM air cooling shall be used to cool the pod internally while in flight. The pods shall be capable of relaying UHF communication signals and Navy Tactical Data System (NTDS) Link 11 signals. System components, as far as possible, shall be Navy inventory items (see figure 1).

3.2 First article. When specified in the contract or purchase order (see 6.2.1), a sample(s) shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Parts and materials. The selection and application of parts, materials, and processes shall be in accordance with MIL-E-5400 and MIL-STD-454, except as specified herein. These requirements shall be applicable to any equipment designed or modified by the contractor or subcontractors, but shall not apply to unmodified off the shelf equipment.

3.3.1 Screening of parts. All electrical, electronic, and electromechanical parts shall be subjected to 100 percent screening. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the

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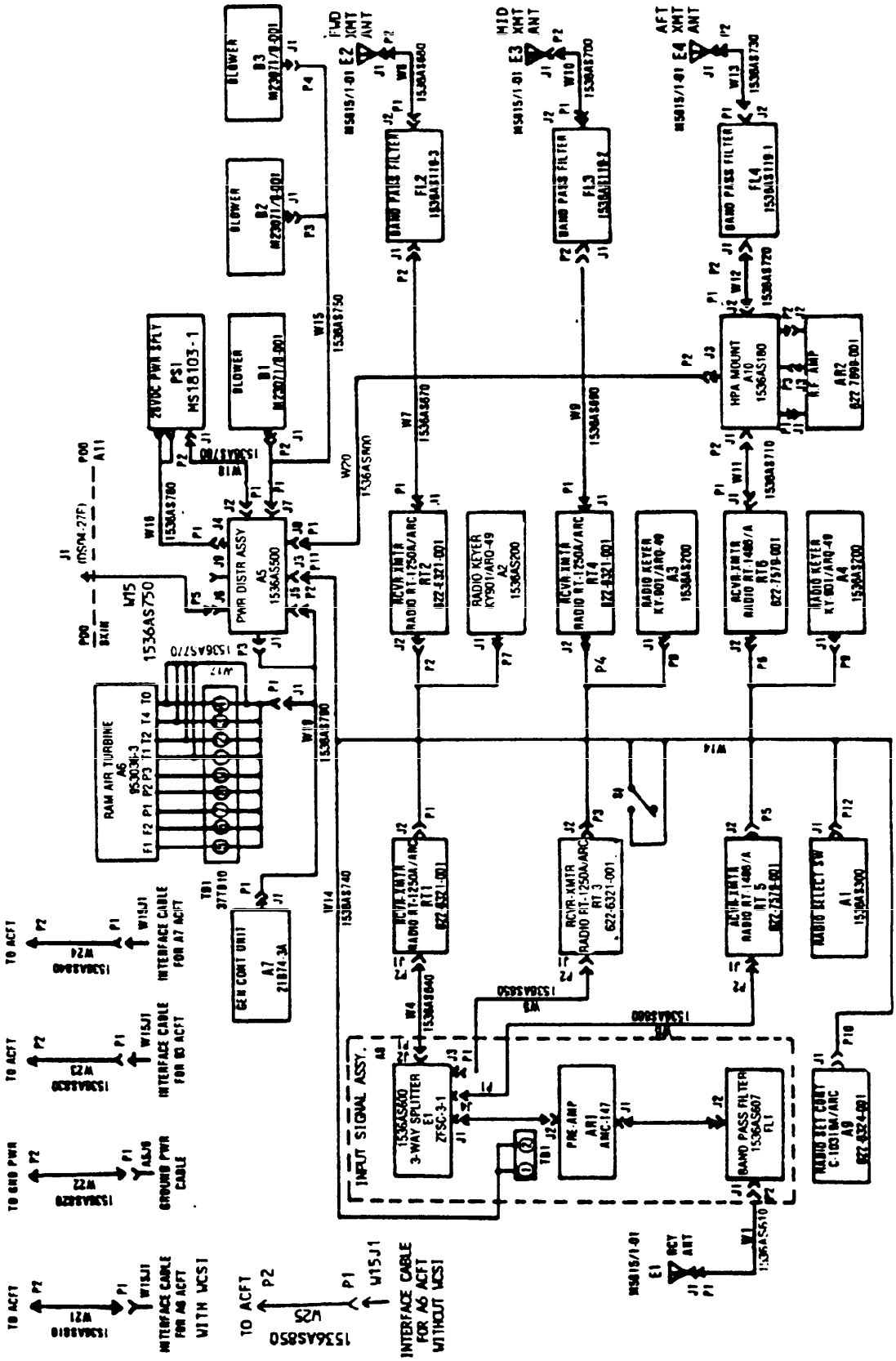


FIGURE 1. Radio Repeater Set Pod AN/ARQ-49, block diagram.

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requirements of the most similar military standard or standard-established reliability part. All integrated circuits shall be screened to MIL-STD-883, Method 5004, Class B as a minimum. Hybrid circuits shall be screened to MIL-STD-883, Method 5008 as a minimum.

3.3.1.1 Incoming inspection. Upon receipt, each active component shall be subjected to a functional test at the ambient temperature. Failed components shall not be used in the pod.

3.3.2 Part derating and application. Application and derating requirements shall be in accordance with AS-4613, Class A. High Power Amplifier (HPA) junction temperatures shall be not greater than 150°C and all other junction temperatures shall be not greater than 110°C.

3.3.3 Stress analysis. The contractor shall conduct a part-level stress analysis to verify derating compliance.

3.3.4 Parts control program. The contractor shall establish and maintain a parts control program in accordance with MIL-STD-965, Procedure I.

3.3.5 Microelectronic modular assemblies. Microelectronic technology shall be considered, and when used, microelectronic items shall meet the requirements of MIL-STD-883. Conformal coatings used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.3.6 Nonrepairable subassemblies. When used, nonrepairable subassemblies shall be in accordance with MIL-E-5400 and MIL-STD-2084.

3.3.7 Metals. Metals shall be of corrosion-resisting type or treated to resist corrosion in salt spray, atmospheric conditions, or fuels during the service life of the pod when in storage or during normal service use.

3.3.7.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact. Dissimilar metals are specified in MIL-STD-889.

3.3.8 Castings. Castings shall be clean and free from blow holes, porosity, cracks and other defects. All castings shall be in accordance with the requirements of MIL-C-6021 and MIL-STD-454, Requirement 21.

3.3.9 Structural welding. Structural welding shall be in accordance with MIL-STD-454, Requirement 13.

3.3.10 Circuit materials. Circuits shall be designed to use a minimum of different materials, parts, subassemblies, cards and modules.

3.3.11 Soldering. The contractor's soldering system procedures and training programs shall be in accordance with the requirements of WS-6536.

3.4 Design. The pod shall be in accordance with MIL-A-8591 and all the applicable requirements of MIL-E-5400, Class 1.

3.4.1 Standard, special tools and test equipment. The pod shall be designed so that standard tools and test equipment or Ground Support Equipment (GSE) used with predecessor equipment may be used to the greatest possible extent to accomplish all necessary adjustments, maintenance and testing.

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3.4.2 Modules. The electronic portions of the pod shall be modularized by function in accordance with MIL-STD-2084, Requirement 103. Mechanical subassemblies shall be modularized to the extent practical to facilitate production assembly and rework and to facilitate changes. The use of cements and welds or other bonds which will consume appreciable rework time shall be avoided in the design of repairable assemblies. Assemblies that require machining, lapping, or custom fitting shall be avoided. Where two units must be matched and used with each other, an assembly drawing shall be created that identifies this requirement. When adjustment or alignments must be made, components shall be pinned, slotted, or scribed to permit rapid reassembly with minimum gauging and trial and error. Replacement of modules shall not require use of solder, cement, or welding. The design shall be such that reverse or improper assembly, mating or installation shall be impossible. Where possible, circuitry loops shall be contained on one Shop Replaceable Assembly (SRA). If the number of discrete components in a loop exceeds the practical or physical space available on one SRA, mounting and packaging shall facilitate maintenance. Non-repairable or disposable subassemblies may be used and are preferred, providing such subassemblies are sufficiently reliable and inexpensive to justify that concept; where significant weight savings may be achieved, or where mechanical integration with or modification of existing design requires encapsulated subassemblies. Items recommended as nonrepairable shall be identified.

3.4.3 Encapsulation. Repairable assemblies shall not be encapsulated, potted, or embedded. When a conformal coating is used, it shall be a transparent type easy to remove and reapply.

3.4.4 Fasteners. The different types of fasteners (screws, bolts, and quick release fasteners) shall be minimized to reduce tool equipments for maintenance. Quick, positive release, self-locking, and captive features shall be used where appropriate.

3.4.5 Cables and connectors. The pod shall be designed for the use of cables and connectors in accordance with MIL-E-5400.

3.4.5.1 Interconnecting cabling. The pod shall be designed to operate as required using the external wiring in accordance with the applicable require ments of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet the interference control requirements, and provide that assembly and maintenance of the cable to the plugs may be easily accomplished. Polyvinylchloride (PVC) insulated conductors or jackets shall not be used.

3.4.5.2 Cable and wire runs. Cable and wire runs shall be routed in such a manner as to avoid sharp bends and pinching, especially near connectors. Protection shall be provided for cables routed through openings. Adequate clamping or support provisions shall be made. Easily removable and replaceable sections of cables shall be used where appropriate.

3.4.6 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location.

3.4.7 Interchangeability. In addition to the requirements of MIL-STD-2084 and MIL-E-5400, all parts having the same part number shall be interchange- able with each other with respect to installation and performance.

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3.4.8 Primary input power. The pod shall be designed to meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as specified in MIL-STD-704 having limits as specified therein. A major design goal of the pod shall be to conserve primary power to the maximum extent possible. The average power required shall be not greater than 1800 watts supplied from 115/208 Vac, 400 Hz, 3 Phase, in accordance with MIL-STD-704.

3.4.8.1 Degraded performance. Degraded performance will be permitted during all voltage transients of MIL-STD-704 for Normal Electric System Operation, except that performance shall not be affected by short pulses of less than 50 microseconds. Operation shall return to normal without requiring reinitialization when the transient expires.

3.4.8.2 Overload protection. Internal overload protection to the pod shall be provided. All parts and circuits of the pod which are likely to experience an overload due to any failures or improper adjustments shall be designed to withstand such overload without damage or shall have protective devices. The use of protective devices shall be held to a minimum.

3.4.8.3 Undervoltage protection. The pod shall not be damaged by voltages below the minimum specified in MIL-STD-704 and shall automatically resume normal operation when the voltage returns within the limits.

3.4.9 Warm-up time. The maximum warm-up time for the pod shall be not greater than 1 minute (see 3.12).

3.4.10 Cooling. Cooling design performance and evaluation as specified in MIL-E-5400 shall be conducted during design and development phase in accordance with MIL-T-23103. The pod shall provide for adequate cooling using ram air during flight and forced ambient (unconditioned) air when on the ground.

3.4.10.1 Thermal protection. Internal thermal protection to the pod shall be provided. All parts and circuits of the pod which are likely to experience a thermal overload due to any failures or improper adjustments shall be designed to withstand such overload without damage or shall have protective devices. The use of protective devices shall be held to a minimum.

3.4.11 Safety. The design of the pod shall incorporate the personnel safety requirements of MIL-STD-454, Requirement 1.

3.4.12 Human performance and human engineering. Human performance and human engineering design criteria for the pod shall be in accordance with MIL-STD-1472. MIL-H-46855 shall be used as a guide.

3.4.13 Power source. Power for operation shall be provided by a front mounted Ram Air Turbine (RAT) or host aircraft while in flight and by ground power while on deck.

3.5 Construction. The pod shall be constructed in accordance with Drawing 1536AS100. The pod shall be constructed in such a manner that the physical and functional relationships between system components are not affected by the strains, impacts, and vibration associated with operation, storage, and shipping. The construction of the pod assembly shall be compatible with

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space limitations and shall be capable of withstanding, without damage, a Class 1 flight environment as specified in MIL-E-5400.

3.6 Equipment characteristics.

3.6.1 Performance characteristics. Values set forth to establish the requirements for performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance required under standard conditions are specified herein.

3.6.1.1 Operating conditions.

3.6.1.1.1 Service conditions. Service conditions shall be as specified in 3.9 and 3.10.

3.6.1.1.2 Airspeed. The system shall be capable of carriage at MACH .85 at 1000 feet above mean sea level.

3.6.1.1.3 Aerodynamic loads. The system shall be capable of withstanding the loads yielded by procedures I and II of MIL-A-8591.

3.6.1.1.4 Mounting. The pod shall be mounted on one of the following Navy aircraft in the position specified in table I.

TABLE I. Aircraft pod mountings.

Aircraft	Position
A-6	Wing station
A-7	Wing station
S-3	Wing station

3.6.1.2 Electrical power. The pod shall operate from the RAT power system in flight or, as an option during preflight, the pod may be configured to operate from aircraft 3Ø power when mounted on the A-6 or A-7. The pod shall require not more than 2 kVA to perform all functions.

3.6.1.2.1 RAT power system. The RAT power system shall be capable of providing adequate power when aircraft speed equals or is greater than 180 knots.

3.6.1.3 Cockpit control. The system shall be activated via a switch in the aircraft cockpit. This switch shall be the only cockpit control for the pod. Existing aircraft wiring shall be used. The activation signal shall not require more than 1/2 amperes at 28 Vdc.

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3.6.1.4 Repeater performance. Repeater performance shall be as follows:

a. Provide three relay channels

(1) Two channels capable of relaying clear or encrypted voice or teletype.

(2) One channel capable of relaying data in link 11 format.

b. Shall not decode or process encrypted information.

c. Provide an operational radius of 200 statute miles when at an operational attitude appropriate for line-of-sight conditions.

3.6.1.4.1 Control. A single remote control unit, C-10319/ARC, shall be used to control all six receiver-transmitter sets. Tuning of the receiver-transmitters shall be accomplished by successively switching the C-10319/ARC from one receiver-transmitter to the next. During flight, the control shall be off.

3.6.1.4.2 Response time. The system shall identify and relay valid input signals as follows:

a. Clear voice 100 ms maximum

b. Encrypted voice 100 ms maximum

c. Link 11 data 15 ms maximum

3.6.1.4.3 Receive frequencies. Radio signals received by the repeater shall be between the frequencies 330.000 MHz and 366.000 MHz.

3.6.1.4.4 Repeater transmit frequencies. Retransmission frequencies shall fall within the following frequency bands:

a. Chan 1: 256.000 MHz to 264.000 MHz

b. Chan 2: 270.000 MHz to 278.000 MHz

c. Chan 3: 284.000 MHz to 292.000 MHz

3.6.1.4.5 Frequency spacing. Each frequency band, both receive and transmit shall be subdivided into discrete channels with 25 kHz spacing.

3.6.1.4.6 Modulation type. Modulation types used on the three repeater bands shall be as follows:

a. Chan 1: Link 11 - frequency modulation (16K0F7D)

b. Chan 2: Clear voice - amplitude modulation (6K00A3E)
- frequency modulation (16K0F3E)

Encrypted voice - amplitude modulation (6K00A3E)
- frequency modulation (16K0F3E)

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c. Chan 3: Clear voice - amplitude modulation (6K00A3E)
- frequency modulation (16K0F3E)

Encrypted voice - amplitude modulation (6K00A3E)
- frequency modulation (16K0F3E)

3.6.1.4.7 Receiver sensitivity. The repeater system shall be capable of providing the relay function with an RF input signal of at least -100 dBm in the AM mode and -100 dBm in the FM mode. Receiver selectivity shall be as follows:

a. RF selectivity. Pod mounted filtering shall provide at least 50 dB rejection to signals that are more than 38 MHz removed from the receive band.

b. Adjacent channel selectivity. Adjacent channel selectivity of the repeater system shall be the same as that afforded by the receiver-transmitter used in the system.

3.6.1.4.8 Transmitted power output. Repeater transmitter power, measured at the antenna, shall be as follows:

a. Chan 1: FM mode not less than 32.5 watts

b. Chan 2: AM mode not less than 7.0 watts
FM mode not less than 10.5 watts

c. Chan 3: AM mode not less than 7.0 watts
FM mode not less than 10.5 watts

3.6.1.4.9 Transmitter spurious output. All transmitter RF outputs shall be less than -90 dBc for frequencies more than 2 MHz outside each transmit band. Spurious outputs within a given transmit band shall be not greater than that allowed for the receiver transmitter used in the system.

3.6.1.4.10 Antenna suitability. All repeater system antennas shall be capable of required operation over the frequency range of 250 to 370 MHz in both the receive and transmit modes.

3.6.1.4.11 Antenna use. The repeater system shall employ one antenna for all radio signal reception and separate antennas for each of the three transmit bands.

3.6.1.4.12 Antenna radiation patterns. All repeater system antennas shall provide radiation patterns which are omnidirectional.

3.6.1.4.13 Antenna impedance. The terminal impedance of all system antennas shall be 50 ohms.

3.6.1.4.14 Antenna standing wave ratio (SWR). Antenna SWR over the frequency range of 250 MHz to 370 MHz shall be not greater than 2 to 1.

3.6.2 Physical characteristics. Physical characteristics of the pod shall be as specified in 3.6.2.1 through 3.6.2.3 and as follows:

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a.	Length:	15 ft ± 1 ft
b.	Diameter:	21 inches ± 1/2 inch
c.	Weight:	Not greater than 600 pounds
d.	Center of gravity:	In accordance with MIL-A-8591 for jettisonable stores

3.6.2.1 Access to equipment. Each of the four bays shall have an access door on the starboard side. Bay 1 and bay 3 shall also have an access door on the port side. Access to equipment in bays 1 and 3 may require both doors to be removed.

3.6.2.2 Access doors. Access doors shall be attached to the pod using lanyards or other attachments.

3.6.2.3 Ejection load. The pod shall be capable of withstanding a ejection load of 25000 pounds applied to the area specified in MIL-A-8591.

3.7 Reliability. The contractor shall establish and conduct a reliability program.

3.7.1 Reliability analysis and predictions. The contractor shall conduct a reliability analysis and predictions report of the system in accordance with MIL-STD-756 and MIL-HDBK-217.

3.7.2 Mean-time-between-failure (MTBF). The system, including Built-In-Test (BIT) provisions, shall have a specified MTBF of not less than 50 hours (0-1) as specified in MIL-STD-781.

3.7.3 Operating life. The pod shall be designed to meet the operational requirements specified in 3.7.3.1 and 3.7.3.2.

3.7.3.1 Total operating life. The pod shall have total operating life of 10 years under the conditions specified in 3.9 and 3.10 with nominal servicing and replacement of parts. Parts requiring scheduled replacement shall be identified by the manufacturer.

3.7.3.2 Totalizing time meter. The pod shall include a time totalizing meter conforming to applicable requirements of MIL-M-7793. The meter shall be located such that it may be read without the removal of any part from the unit. The meter shall indicate "ON" time for any operating mode, including standby, shall have a range from 0 to 9,999 hours, and shall not be capable of being reset.

3.8 Maintainability. The contractor shall incorporate the design requirements for maintainability features, BIT features, construction, provisions for test points, and other maintainability parameters in accordance with MIL-STD-2084 as modified herein.

3.8.1 General. The system shall be an amalgamation of existing equipments; however, within that constraint, the design, configuration and installation of the pod shall minimize maintenance or support requirements, including the need for special tools and support equipment, personnel skills, manpower, elapsed maintenance time and total maintenance time at all levels of maintenance. All maintenance tasks shall be capable of being performed at

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the organizational, intermediate, and depot maintenance levels. The design shall be such that the maximum number of maintenance tasks may be accomplished at the organizational and intermediate maintenance levels. Requirements for adjustments of the SRA at the intermediate and depot level shall be necessary only following replacement due to failure.

3.8.2 Remove or replace provisions. Design of the pod shall make provisions for rapid removal and replacement of each SRA in the pod with no requirement to remove or displace other replaceable items, parts, or subassemblies.

3.8.3 Built-in-test (BIT). BIT of the receiver or transmitters shall be provided by the AN/ARC-182 Remote Control.

3.8.4 Test points and test features. Suitable test points and test features shall be included in the design of the pod. Selection and accessibility of test points shall be compatible with the design of replaceable assemblies and the level of maintenance at which the test points are to be used.

3.8.5 Mean time to repair (MTTR). The MTTR of the pod shall be not greater than 60 minutes at the organizational and intermediate levels.

3.9 Environmental conditions. The pod shall operate as required under the conditions specified herein.

3.9.1 Standard conditions. The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the pod:

a. Temperature	Room ambient ($25 \pm 5^{o}C$)
b. Altitude	Normal ground (MSL to 6000 feet)
c. Vibration	None
d. Humidity	Room ambient up to 90 percent relative humidity
e. Input Voltage	115 +3, -7 volt; 400 ± 7 Hz, 3 Phase, in accordance with MIL-STD-704

3.9.2 Service conditions. The pod shall be designed to operate as required under any of the environmental service conditions or combination of the conditions as specified in MIL-E-5400 for Class 1 equipment.

3.9.2.1 Temperature-altitude. The pod shall be designed to meet the conditions described in the temperature-altitude test requirements of MIL-T-5422 for Class 1 equipment except the altitude shall be not greater than 50,000 feet (see 4.6.1).

3.10 Electromagnetic interference. The pod shall comply with the requirements of MIL-STD-461 for Class A1b equipment except as modified herein. The requirements shall be met in all operating modes and at all operating frequencies of the radios.

3.10.1 Conducted and radiated emission limits. The following conducted and radiated emission limits of MIL-STD-461, Part 2 apply:

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a. CE03 - Except that the limits shall be a curve everywhere representing a 20 dB relaxation from the limits shown in MIL-STD-461, figures 2-2 and 2-3.

b. CE06 -

c. RE02 - Except that the limit shall be a curve everywhere representing a 20 dB relaxation from the No. 2 curves of MIL-STD-461, figures 2-9 and 2-10.

3.10.2 Conducted and radiated susceptibility limits. The following conducted and radiated susceptibility limits of MIL-STD-461 shall apply:

a. CS02 - Except that this requirement shall apply to all neutrals and grounds whether or not they are internally returned to chassis ground.

b. CS03 - The upper frequency limit shall be 1500 MHz.

c. CS04 - The upper frequency limit of the standard (CW) CS04 interfering signal shall be 4 GHz.

d. CS06 -

e. RS03 - The field levels and modulations shall be as specified in table II of this specification.

3.11 Identification and marking.

3.11.1 General. All markings shall be durable to prevent effacing or obliteration resulting from service usage.

3.11.2 Marking of parts. Each part of assembly shall be marked in accordance with MIL-STD-130 and with the part number which shall be the same as the drawing number of that part or assembly. An assembly consists of parts that are permanently fastened together by welding, brazing, soldering, or riveting. Exceptions to this are parts not having suitable or sufficient surface for a part number.

3.11.3 Location of part number. The part number shall, when practicable, be located to permit its being read after assembly in the complete unit.

3.11.4 Identification plate. The identification plate for the pod shall conform to MIL-P-15024/2, Type H, and shall be located so that it is readily visible. The identification plate shall provide the following information:

a. AN/ARQ-49

b. Contract or Purchase Order Number

c. Drawing Number

d. Manufacturer's Code Identification

e. Manufacturer's Serial Number

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TABLE II. RS03 signals.

Frequency (MHz) 7/	Modulation	Power Density (mW/cm^2) Peak 5/	Power Density (mW/cm^2) Ave	V/M (Ref) 6/
0.014-850	AM 90% 1 KHz Sine Wave	-	.106	20
215-225	PAM 1/ 20 usec 2/ PW 3/ 200 pps 4/	.317	-	35
215-225	PAM 200 usec PW 200 pps	.317	-	35
400-450	PAM 15 usec PW 300 pps	.106	-	20
850-940	PAM 10 usec PW 1000 pps	.106	-	20
850-940	PAM 125 usec PW 300 pps	.106	-	20
1215-1365	PAM 10 usec PW 3000 pps	.106	-	20
1365-2900	PAM 1 usec PW 1000 pps	.106	-	20
2900-3100	PAM 3 usec PW 1000 pps	1.6	-	78
2900-3100	PAM 70 usec PW 200 pps	1.6	-	78
3100-9000	PAM 1 usec PW 1000 pps	.106	-	20
9000-14000	PAM 0.2 usec PW 1000 pps	.106	-	20
9000-14000	PAM 0.2 usec PW 1000 pps	.106	-	20
14000-18000	PAM 0.2 usec PW 50000 pps	.106	-	20

1/ Pulse Amplitude Modulated.

2/ Microseconds.

3/ Pulse Width.

4/ Pulses Per Second.

5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; see 4.9.7 of MIL-STD-461.

6/ Volts Per Meter equivalent shown for reference only.

7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and/or field strength shown.

3.12 Finish. Protective coatings and finishes which will crack, chip, scale, during normal service life or due to extremes of atmospheric conditions, shall not be used.

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3.12.1 Aluminum alloy. Chemical Film, MIL-C-5541, Class 3 shall be applied to all aluminum surfaces. The surfaces shall meet the paint adhesion requirements of MIL-C-5541.

3.12.2 Carbon and alloy steel. All steel parts shall be cadmium plated in accordance with QQ-P-416.

3.12.3 Corrosion resistant steel. Corrosion resistent steel shall be passivated in accordance with QQ-P-35.

3.12.4 External finish. Epoxy primer, MIL-P-23377, Type II and polyurethane coating, MIL-C-83286 color number 36375 shall be applied in accordance with MIL-F-18264.

3.13 Bonding. Electrical bonding of the installed equipments to the pod shall comply to the requirements of MIL-B-5087, Class A, H, R and S. Bonding resistance between the bonding surface and all parts of the pod shall be not greater than 2.5 milliohms.

3.14 Workmanship. Workmanship for the pod shall conform to MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in standard conditions (see 3.9.1).

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4.3.1 Performance characteristics. During inspection acceptance testing, the equipment shall meet the requirements of Section 3.

4.4 First article inspection. First article inspection shall be conducted by the contractor on one pod representative of the production pods to be supplied under the contract. First article tests shall be accomplished using the authorized test procedures. Column I of table II lists minimum tests required for first article inspection. No first article inspections shall be conducted prior to procuring activity acceptance of the first article test procedures.

TABLE III. First article and quality conformance inspection.

Minimum Test Requirements	I. First Article Inspection	II. Quality Conformance Inspection	
		Individual	Special
Visual Examination (3.4 through 3.4.7)	4.5.1.1	4.5.1.1	4.5.1.1
Performance Characteristics (3.11)	4.5.1.2	4.5.1.2	4.5.1.2
Workmanship (3.16)	4.5.1.3	4.5.1.3	4.5.1.3
Environmental Tests (3.9)	4.6	1/	2/
Warm-up time (3.4.9)			
Interference Control (3.10)	4.7		
Reliability (3.7)		4.5.1.4	

1/ Additional individual tests may be requested by the procuring activity (see 4.5.1.4).

2/ Special tests to be performed will be determined by the procuring activity (see 4.5.2.1.b).

4.4.1 First article acceptance. Acceptance of the first article sample will be by the procuring activity upon successful completion of all tests (see 6.2.2). No production pods shall be delivered prior to the acceptance of the first article sample. Prefabrication of production equipment prior to the acceptance of the first article sample is at the contractor's own risk.

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4.4.2 Production equipments. Pods supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equal to the accepted first article sample. Each pod shall be capable of successfully passing the same tests as imposed on the first article sample.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests (see 6.2.2). All inspection and testing will be under the supervision of the Government inspector. Quality conformance tests shall consist of the following (see table III, column II):

a. Individual tests (see 4.5.1)

b. Special tests (see 4.5.2)

4.5.1 Individual tests. Each pod submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each pod shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

c. Manufacture screening (see 4.5.1.3)

d. All equipment reliability assurance test (see 4.5.1.4)

4.5.1.1 Examination of product. Each pod shall be examined carefully to determine that the material, interchangeability, workmanship and safety requirements have been met. Particular attention shall be given to the following:

a. Completeness

b. Nameplate, identification marking, and labels

c. Alignment and tightness

d. Finish

e. Fit and placement of components

f. Ease of operation of jacks, switches, sliding parts and controls

g. Mountings and brackets

h. Fastening and securing of devices or parts

i. Accessibility of components and parts

j. Welded and soldered joints

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k. Cable runs between components including plugs and receptacles

l. Forms, harnesses, and other wiring

m. Grounding connections

n. Weight

o. Overall dimensions

p. Potential shorts and defective insulation

q. Foreign objects, lint, dust

r. Other visual defects

s. Workmanship

t. Interchangeability

4.5.1.2 Operational tests. Each pod shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure specified equipment operation at room ambient in accordance with the accepted test procedure (see 4.4).

4.5.1.3 Manufacturing screening.

4.5.1.3.1 WRA level. Components of the pod, whether to be installed in pods or furnished as spares, shall be subjected to manufacturing screening and other reliability testing as specified in the individual drawings and specifications for the equipments. In addition, the Radio Keyer Assemblies, 1536AS200; R/T Switch Assemblies, 1536AS300; and the Power Distribution Assemblies, 1536AS500 shall be subjected to manufacturing screening as specified in 4.5.1.3.1.1 and 4.5.1.3.1.2.

4.5.1.3.1.1 Random vibration. Each equipment shall be subjected to 10 minutes of random vibration according to the spectrum shown on figure 2. The vibration shall be applied in the direction perpendicular to the majority of the circuit boards.

4.5.1.3.1.2 Temperature cycling. Equipments to be furnished as spares shall undergo nine temperature cycles consisting of reducing the temperature to -54°C until the temperature stabilizes, then increasing the temperature to +100°C at a chamber temperature change rate of at least 5°C/minute and maintaining +100°C until equipment temperature stablizes. The equipments shall be unenergized during cycling. The cycle is shown on figure 3. After the ninth cycle a performance test shall be performed on each equipment at -54°C and +71°C.

4.5.1.3.2 Pod level.

4.5.1.3.2.1 Random vibration. Each pod shall be subjected to 30 minutes of random vibration in accordance with the spectrum shown on figure 4. The pod shall be energized and performance tested during vibration. Functional vibration shall be performed in the vertical axis in accordance with levels shown on figure 4.

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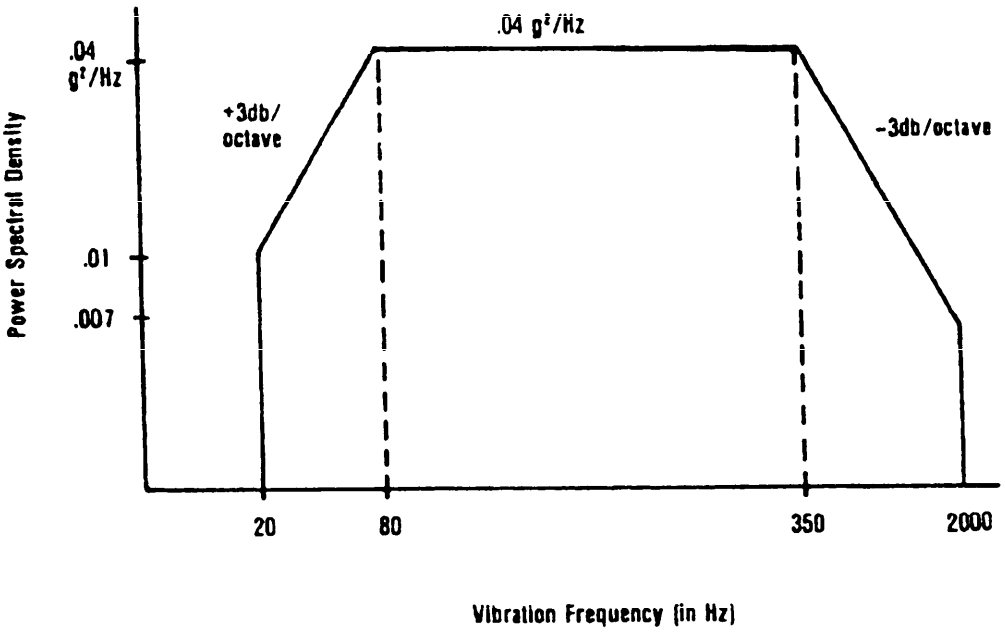


FIGURE 2. Random vibration spectrum.

4.5.1.3.2.2 Temperature cycling. Each pod shall be subjected to temperature cycling in accordance with the environmental cycle shown on figure 3. Temperature cycling shall consist of reducing the temperature with equipment de-energized to -54°C at a minimum rate of 5°C/minute and held until the equipment temperature is stabilized. The units shall then be turned on and a performance check made. The chamber temperature shall then be raised with equipment powered up to +55°C at a minimum rate of 5°C/minute and held until the equipment temperature stabilizes. A performance test shall then be performed on the pod. This procedure shall be repeated until a minimum of not less than 10 cycles are completed with the last cycle failure free.

4.5.1.4 All equipment reliability assurance test. Each pod shall be tested in accordance with test plan XVIIIC of MIL-STD-781 as modified herein. The test cycle shall be in accordance with the environmental cycle of figure 3. Each pod shall be tested for a period of 25 operating hours (5 cycles) with the last cycle failure free. Accept/reject shall be based on an upper test MTBF of 60 hours and a lower test MTBF of 20 hours. Any time a reject situation is reached, the procuring activity reserves the right to stop acceptance of pods pending a review of the contractors efforts to improve the pods, the pod parts or workmanship so that a reject situation can be overcome.

4.5.2 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the pod assembly and to assure adequate quality control. The pod selected for special tests may be selected from pods previously subjected to other tests.

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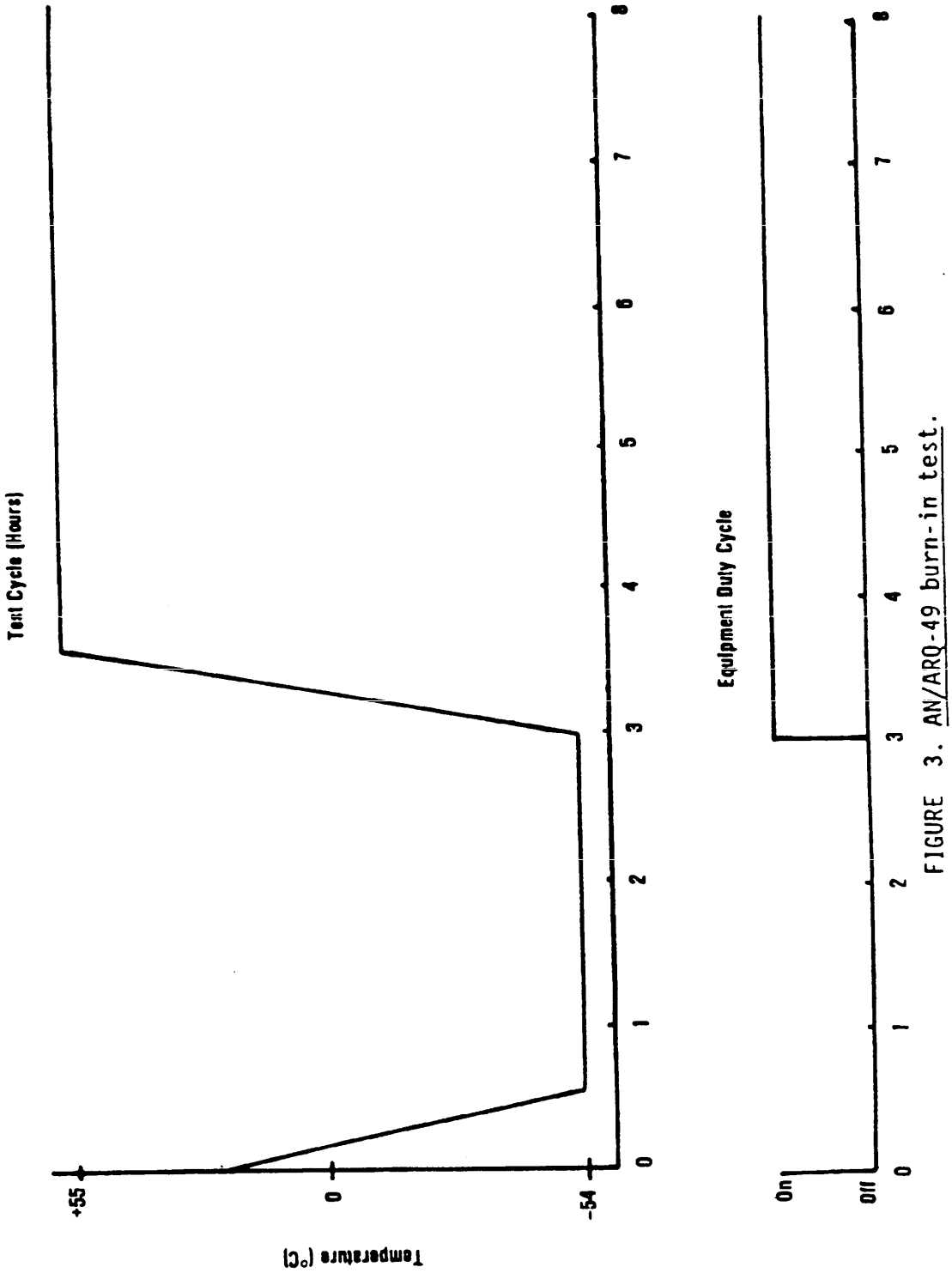
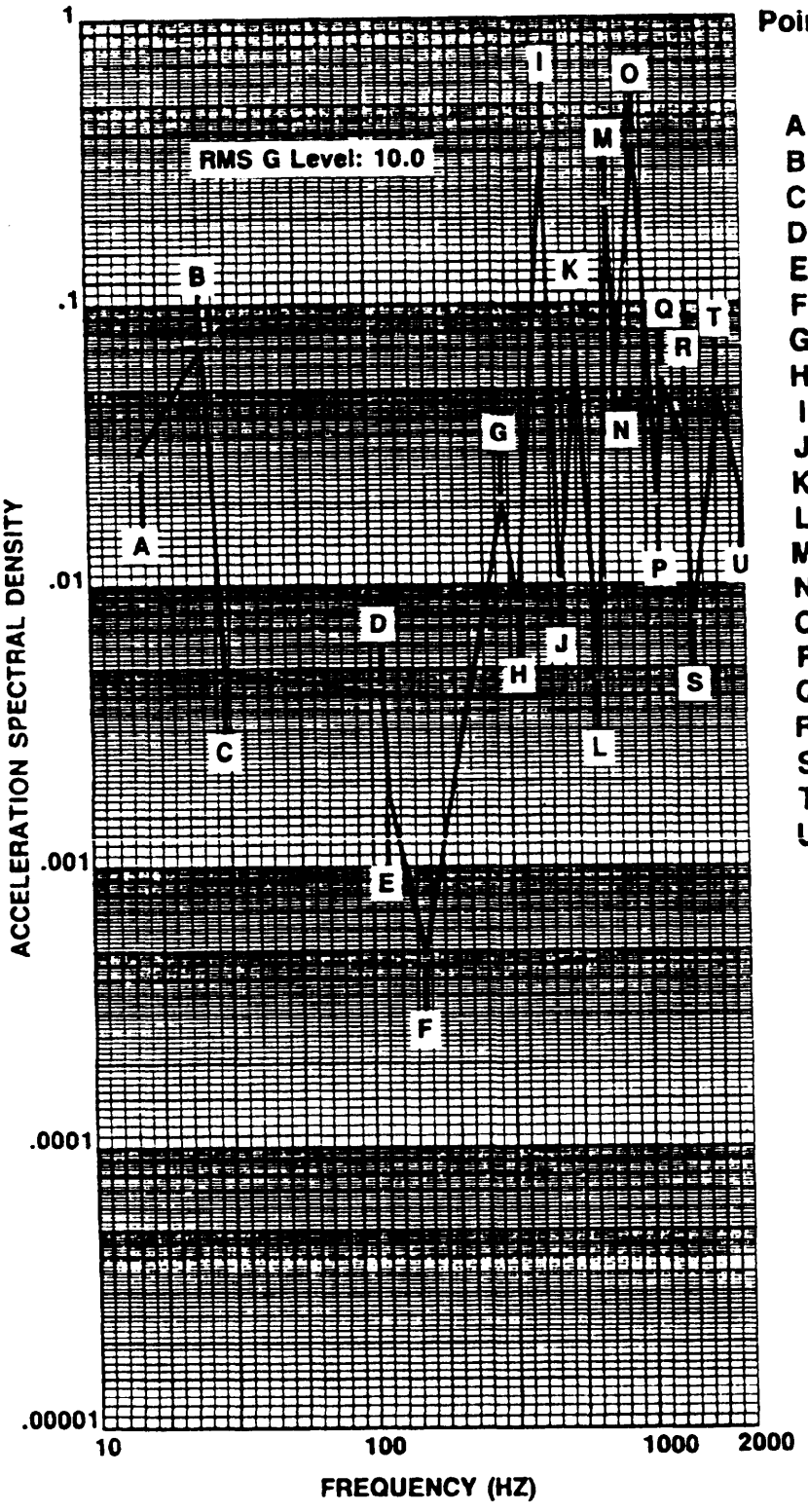


FIGURE 3. AN/ARQ-49 burn-in test.

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Point	Frequency (Hz)	Response Level (g^2/Hz)
A	16	.03
B	24	.07
C	32	.005
D	104	.004
E	112	.002
F	144	.0005
G	296	.019
H	320	.009
I	400	.36
J	456	.011
K	504	.072
L	616	.005
M	696	.22
N	720	.064
O	832	.36
P	1000	.023
Q	1080	.053
R	1304	.033
S	1368	.008
T	1640	.048
U	2000	.023

FIGURE 4. Functional level test spectrum (vertical axis).

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4.5.2.1 Special test schedule. Selection of pods for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.2.2 Scope of tests. Special tests shall consist of such as tests as authorized by the procuring activity. Test procedures previously approved shall be used where applicable.

4.5.3 Manufacturing burn-in and screening. The manufacturing burn-in and screening program shall be conducted in accordance with NAC R&M-STD-R00301.1 to expose deficiencies and defects. Every deliverable unit shall be subjected to the burn-in and screening sequence. Unit interface will be monitored during the burn-in and screening cycle to determine functional operational status of the units. This procedure shall consist of the following sequence of events and shall be conducted in accordance with figure 5.

a. Random vibration

b. Temperature cycling

c. Acceptance test

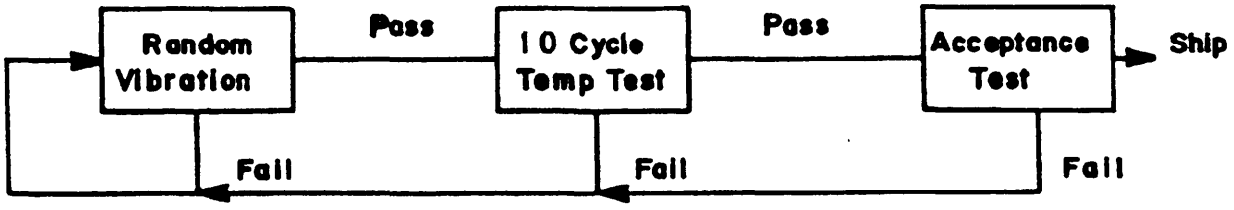


FIGURE 5. Burn-in and screening test sequence.

4.5.3.1 Random vibration. The units shall be subjected to the random vibration spectrum in accordance with figure 2. This random vibration exposure shall be 10 minutes in the axis perpendicular to the majority of the printed circuit boards. The units shall be operating during this time.

4.5.3.2 Temperature cycling. Prior to acceptance, each unit shall successfully be subjected to 10 temperature cycles with the last cycle failure free (see figure 3) as follows:

a. Step 1. The chamber air temperature shall be reduced at a rate of not less than $5^{o}C$/minute and the units stabilized at $-54^{o}C$.

b. Step 2. The units shall than be turned "on" and the chamber air temperature increased at a rate of not less than $5^{o}C$/minute but not greater than $10^{o}C$/ minute from $-54^{o}C$ to $+55^{o}C$ until the units' operating temperature has stabilized (units' temperature change of less than $2^{o}C$ per hour). The units shall be monitored functionally during this time.

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c. Step 3. The units shall then be turned "off". The chamber air temperature shall be reduced at a rate of not less than 5°C/minute and the units stabilized at -54°C. Step 2 through 3 shall be repeated until five consecutive failure-free cycles have been completed.

d. Step 4. At the end of failure-free cycle 5, the units shall be turned "off" and the chamber temperature returned to ambient. In the event of a failure, the five cycles shall be started over.

4.6 Environmental tests. The pod shall operate as required when subjected to the tests of 4.6.1 through 4.6.4.

4.6.1 Temperature-altitude. The pod shall be subjected to temperature-altitude combination requirements of MIL-T-5422 for Class 1 equipment.

4.6.2 Vibration. The pod shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.3, Category 7A, Test Procedure IV.

4.6.3 Shock. The pod shall be subjected to shock tests in accordance with MIL-STD-810, Method 516.3, Procedure IX.

4.6.4 Humidity. The pod shall be subjected to humidity tests in accordance with MIL-STD-810, Method 507.2, Procedure III.

4.7 Electromagnetic interference (EMI) tests. The requirements of 3.10 through 3.10.2 shall be verified by test. Test methods of MIL-STD-461, with the modifications and additions herein, shall be used to assure compliance with the requirements of MIL-STD-461 for Class A1b equipment.

4.7.1 Conducted emissions, power and interconnecting leads, 0.015 to 50 MHz (CE03). Conducted emissions measurements shall be made over the frequency range of 0.015 to 50 MHz on power and interconnecting leads in accordance with MIL-STD-462, Test Method CE03, for power leads and test method CE04 for interconnecting leads.

4.7.2 Conducted emissions, antenna terminals 10 KHz to 26 GHz (CE06). Conducted emission tests shall be performed at the antenna terminals over the frequency range of 10 kHz to 18 GHz in accordance with MIL-STD-462, Test Method CE06.

4.7.3 Radiated emissions, electric field, 14 KHz to 10 GHz (RE02). Radiated emission emanating form the pod shall be measured over the frequency range of 14 kHz to 10 GHz in accordance with MIL-STD-462, Test Method RE02.

4.7.4 Conducted susceptibility, power leads, 30 Hz to 50 kHz (CS01). The pod shall be subjected to conducted susceptibility tests by injecting signals over the frequency range of 30 Hz to 50 kHz on power leads in accordance with MIL-STD-462, Test Method CS01. The pod shall be monitored for malfunction or degradation of performance during the test.

4.7.5 Conducted susceptibility, power leads, 0.05 to 400 MHz (CS02). The pod shall be subjected to conducted susceptibility tests by injecting signals over the frequency range of 0.05 to 400 MHz on power leads in accordance with MIL-STD-462, Test Method CS02. The pod shall be monitored for malfunction or degradation of performance during the test.

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4.7.6 Intermodulation, 15 kHz to 10 GHz (CS03). The pod shall be subjected to an intermodulation susceptibility test in accordance with MIL-STD-462, Test Method CS03. Measurement of the susceptibility characteristics shall be performed during the test.

4.7.7 Conducted susceptibility, 30 Hz to 10 GHz rejection of undesired signals at input terminals (2-signal generator method) (CS04). The pod shall be subjected to a conducted susceptibility test to determine the spurious response of the pod receivers. The test shall be conducted in accordance with MIL-STD-462, Test Method CS04.

4.7.8 Conducted susceptibility, spikes, power leads (CS06). The pod shall be subjected to conducted susceptibility tests consisting of imposing a spike on the power line voltage waveform in accordance with MIL-STD-462, Test Method CS06.

4.7.9 Radiated emissions, electric field, 14 KHz to 12.5 GHz (RS03). Radiated emission emanating from the pod shall be measured over the frequency range of 14 KHz to 18 GHz in accordance with MIL-STD-461, Test Method RS03, and table I of this specification.

4.8 Test procedures. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

4.9 Equipment failure. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074. Should a failure occur during either the individual or special tests, the following action shall be taken:

a. Determine the cause of failure and the equipment BIT performance in detecting and isolating the failure.

b. Determine if the failure is an isolated case or design defect.

c. Where practical, include a test in the individual test to check all pods for this requirement until assurance is obtained that the defect has been corrected.

4.10 Presubmission testing. No item, part, or complete pod shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. Unless otherwise specified in the contract or purchase order, all major units and parts of the pod shall be preserved, packaged, packed, and marked for the level of shipment specified in MIL-STD-794 (see 6.2.1). In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794 and Method 11A of MIL-P-116.

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6. NOTES.

6.1 Intended use. The Repeater Set, Radio, Pod AN/ARQ-49 is intended for use on A6, A7, and S3 aircraft to relay UHF communication signals and Navy Tactical Data System (NTDS) Link 11 signals.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. Selection of applicable levels of preservation packaging and packing (see 5.1)

c. When a first article inspection is required for inspection and approval (see 3.2)

d. If test procedures are to be expanded or otherwise modified (see 4.8)

e. If reliability tests are to be performed (see 4.1.4)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID No.	Option
3.3.3	Report, Reliability Stress Analysis	UDI-R-21423	-----
3.7, 3.7.1	Reliability Prediction and Documentation of System	DI-R-7095	-----
4.4, 4.8	Procedure, First Article Inspection	DI-T-4901	----
4.4.1	Report, First Article Inspection	DI-T-4902	----
4.4.2	Report, Production	DI-T-4904	
4.8, 4.5.2.2	Procedures, Production/ Acceptance Inspection	DI-T-4903	----

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4.9	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719	----
4.9	Report, Failure	DI-R-4805	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior to Government approval is presently appropriate for the pending contract. The first article unit may be refurbished and shipped to satisfy the contractual requirements after successful completion of all required testing.

6.4 Subject term (key word) listing.

A6, A7, S3 Aircraft
Access doors
Aircraft interface
Antenna
Built-in-test
Clear voice
Encrypted voice
Generator control unit
High power amplifier
Link 11
Navy tactical data system
Radio
Ram air turubine
Receiver-Transmitters
Repeater set
Tactical data systems
UHF
Wing station

Preparing Activity:
Navy AS
(Project 1430-135)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-85669(AS)	Repeater Set, Radio Pod AN/ARQ-49, Specification for

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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